UMass Boston

Plan B, The Show Must Go On

Check the fact

Pick only what fits

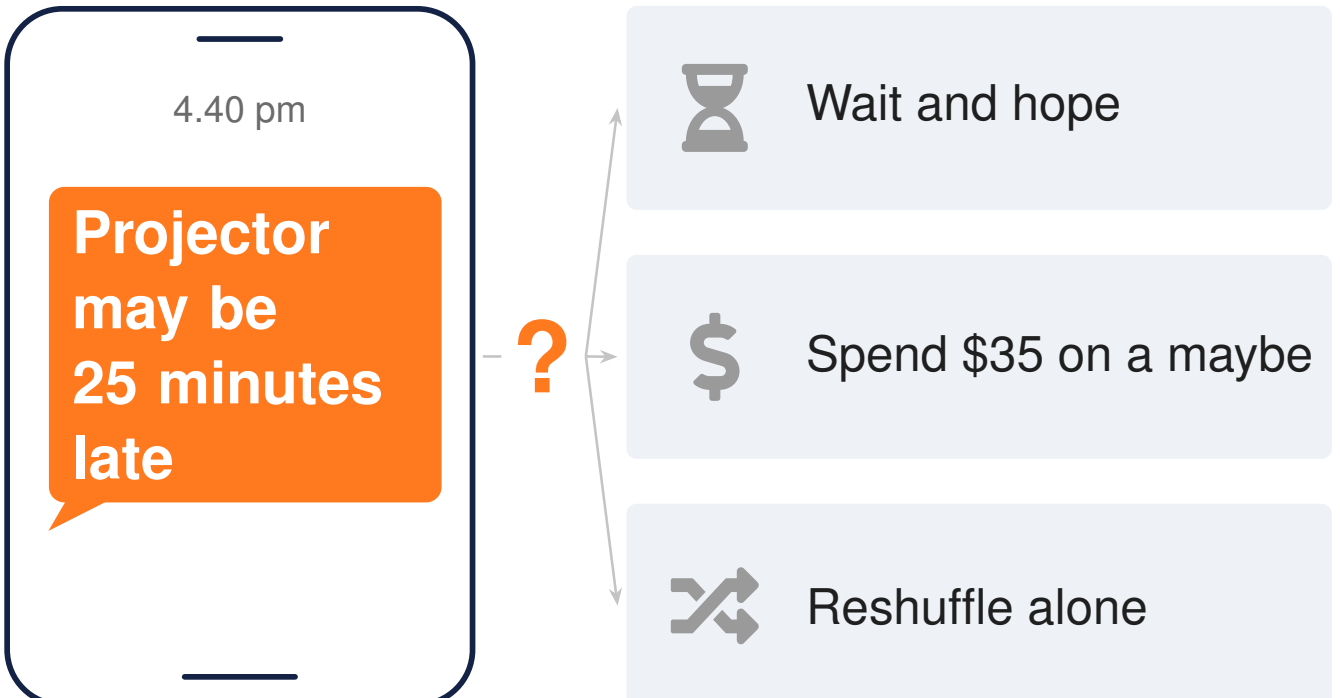
Leave a receipt

Noctis Zhang[1], Zhen Lu[1] and Yinzhi Jin[2]

[1]Department of Computer Science, University of Massachusetts Boston
[2]Department of Management Information Systems, Carnegie Mellon University

PEAAII AI Applications Hackathon 2026, October 2026

Twenty minutes before doors open, Jordan gets a text



Jordan, a volunteer organizer we made up.

Plan B rests on three simple ideas

1



Check the fact

1 look-up, 2 minutes

2



Pick only what fits

16 changes, 3 fit

3



Leave a receipt

4.43 pm, $0 spent

Money waits for a person's OK

First, Plan B checks the one fact that matters

25 minutes late?



looks up the delivery

Delayed, about 5.25 pm

False alarm

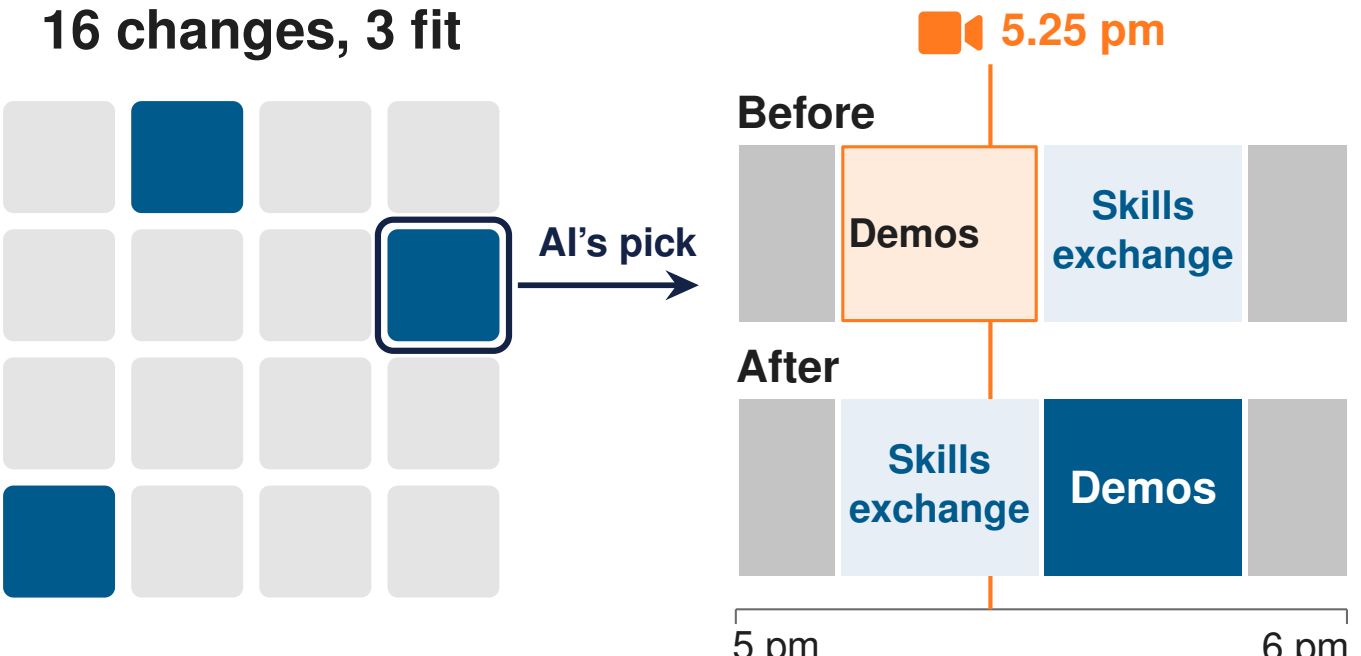
Other order

Repeat

→ **Plan kept**

A message is a claim, not a fact.

THEN THE AI PICKS ONLY FROM CHANGES THAT FIT



The AI chooses. Plan B makes the change.

Every change leaves a receipt, and money waits for a person's OK

Receipt

4.43 pm

Swap demos and skills exchange

Version 1 to 2

$0 spent, $35 not booked



Doors open on time

THIS IS THE REAL BOARD AFTER THE SWAP

Before and after

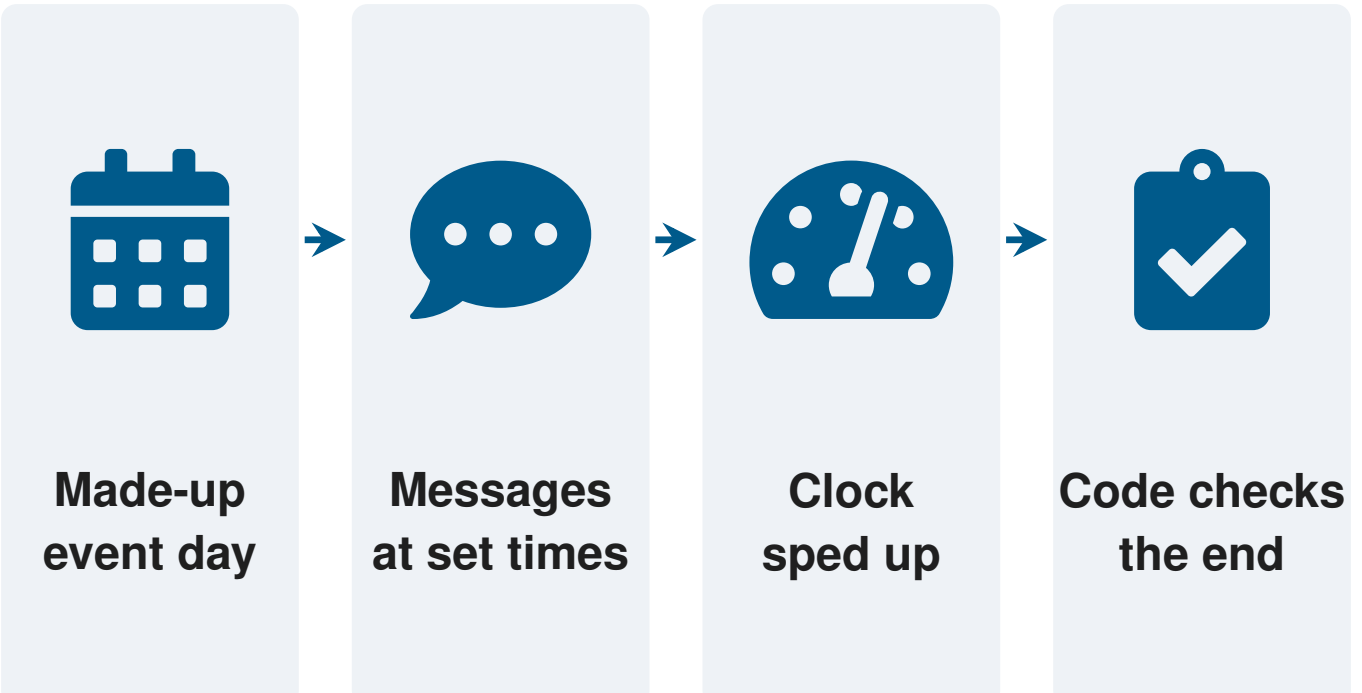
ACTIVITY	BEFORE (V1)	AFTER (V2)
Prepare the event	16:30–16:50 Fixed community room	16:30–16:50 Fixed community room
Welcome	17:00–17:10 Fixed community room, Event host	17:00–17:10 Fixed community room, Event host
Community demonstrations	17:10–17:30 Fixed community room, Primary projector delivery, Demonstration lead	17:30–17:50 Fixed community room, Primary projector delivery, Demonstration lead
Skills exchange	17:30–17:50 Fixed community room, Skills facilitator	17:10–17:30 Fixed community room, Skills facilitator
Closing	17:50–18:00 Fixed community room, Event host	17:50–18:00 Fixed community room, Event host

Everything else unchanged

Two activities swapped

The running board, in a rehearsal with the fixed rules and no AI.

We tested Plan B on event days it had never seen



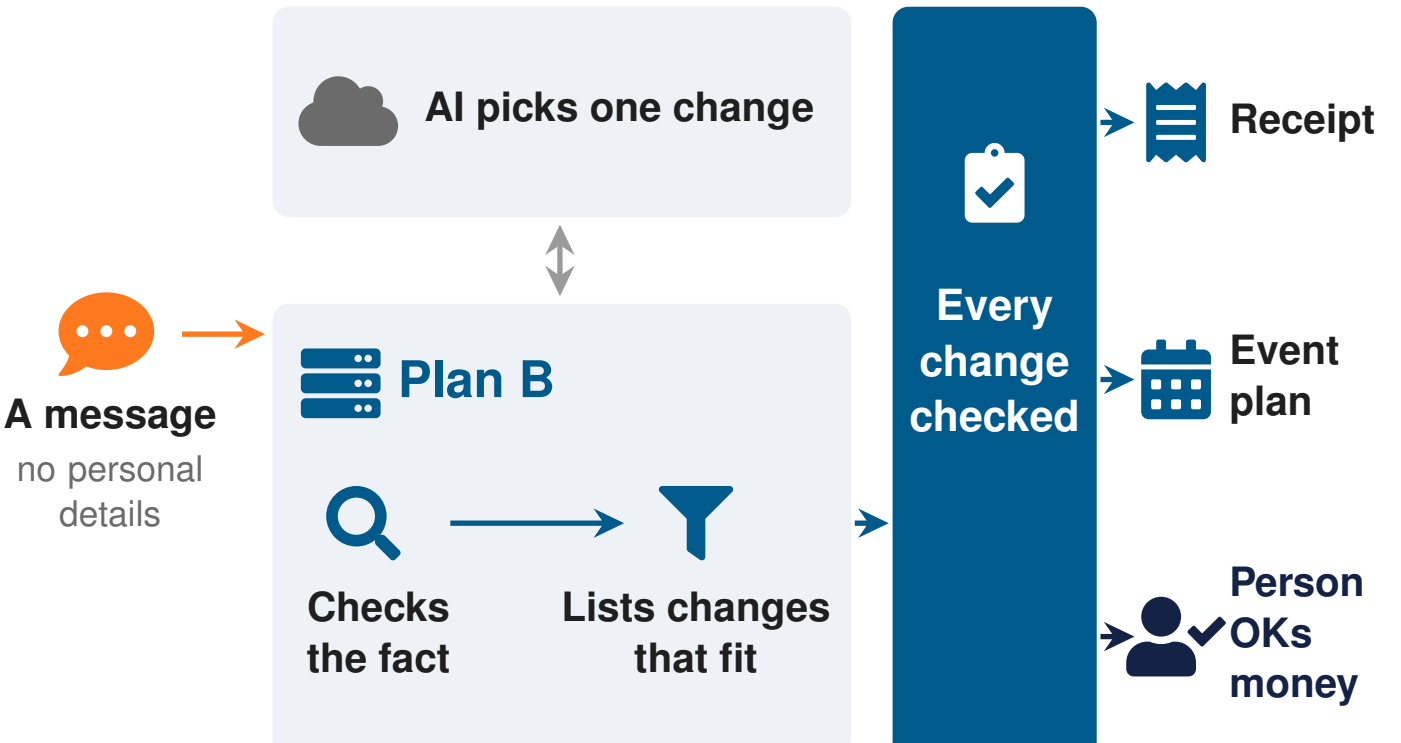
Nothing real is booked or bought.

59 OF 60 RUNS MET THE EXPECTED OUTCOME



Every AI run met it. No executed action broke a rule.

Inside Plan B, careful with messages and money



Built for student clubs and volunteer groups



Club officers

100+ UMass Boston groups[28]



Volunteer coordinators

A helper runs late



Pantry staff

20 of 20 runs met

made-up pantry days

It checks the text before you spend club money.

Other tools help, and Plan B works beside them

Indirect, today

- Group chat[11]
- Shared sheet[12]
- Sign-up sheet[17]
- Campus portal[15]

Direct, change plans

- Sessionboard[1]
- Cvent[2]
- Bizzabo[3]
- AI assistants[4,5,6]

Plan B

From their own pages.

Plan B is safer, more flexible and more personal



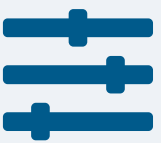
Safer

The three ideas, built in



More flexible

3 kinds of event, your AI or none



More personal

Your event, your spending limit

We have not tested the others.

Check the fact, pick only what fits, leave a receipt

1



One look-up

2



The free swap

3



Doors on time, $0

Money waits for a person’s OK

Code
github.com/nocdoggo/FA26-COLLIE

Next, a walk-through with staff.

Thanks, PEAAII organizers

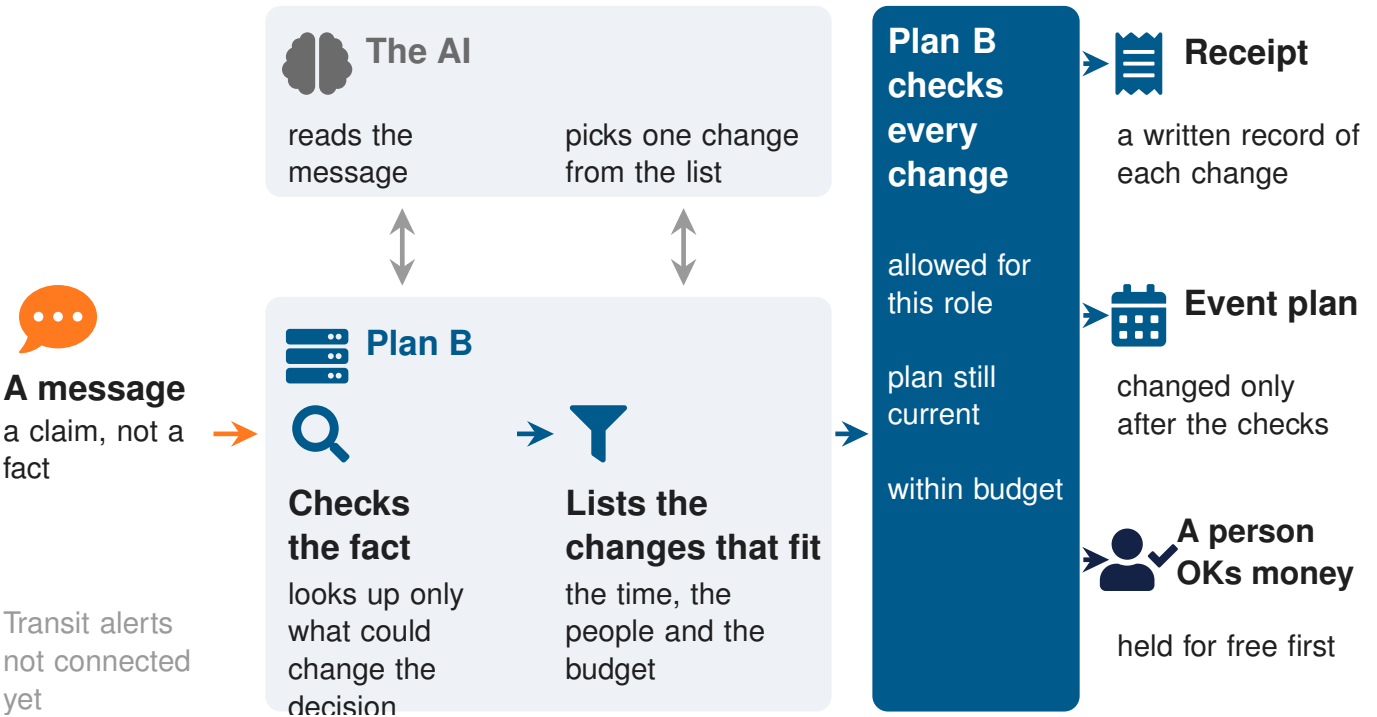
More detail, the six sealed test cases

Case	What happens	Met means	No AI	With AI
Projector late	The projector is really late, past its slot, so the swap does not work	a ten minute shift, or the spare after a person's OK	1 of 1	9 of 9
False alarm	An anonymous call says the projector will be late. The delivery is on time	no change at all	1 of 1	9 of 9
Volunteer late	The skills exchange leader texts that their bus is out of service	the swap, so every activity still happens	**0 of 1**	9 of 9
Cannot be saved	The projector fails its test, and the spare cannot be confirmed in time	Plan B says plainly it cannot be fully done	1 of 1	9 of 9
Pantry delivery	A late delivery cannot arrive in time, and a partner's kits can	40 kits handed out by the deadline	1 of 1	9 of 9
Pantry claims	Wrong and stale delivery claims, and a partner who declines	only kits really on the shelf counted, the shortfall said plainly	1 of 1	9 of 9

Each case ran once with the fixed rules and three times with each of three AI approaches (our coordinator, the check-all variant, the tool-calling model). The fixed rules were never written to read the volunteer's own message. In that case no approach looked anything up, all took the volunteer's word. The six cases use new wording but the same kinds of trouble as earlier cases.

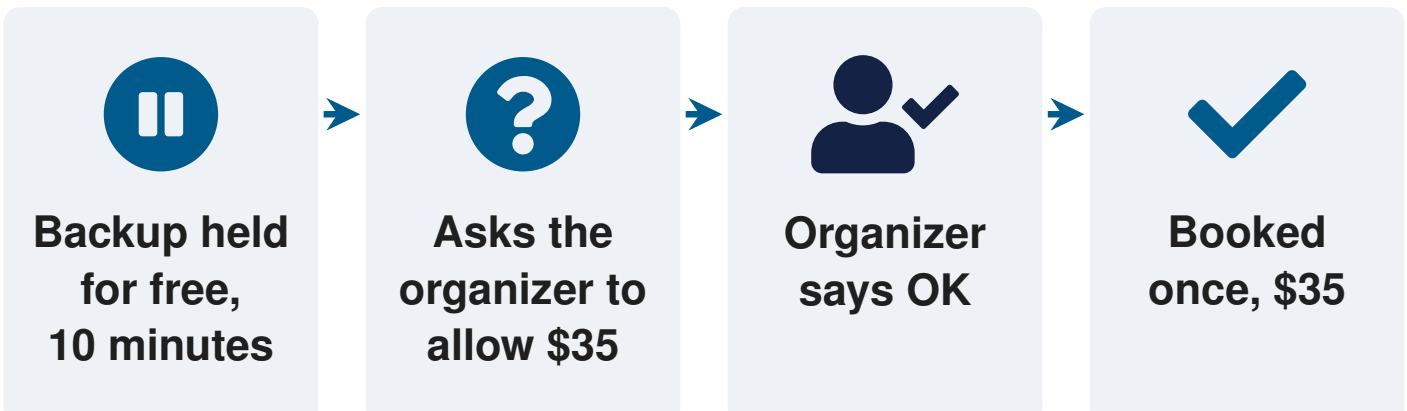
Source, the final test's results folder in the code repository, with its notes on the one miss.

More detail, how Plan B works inside



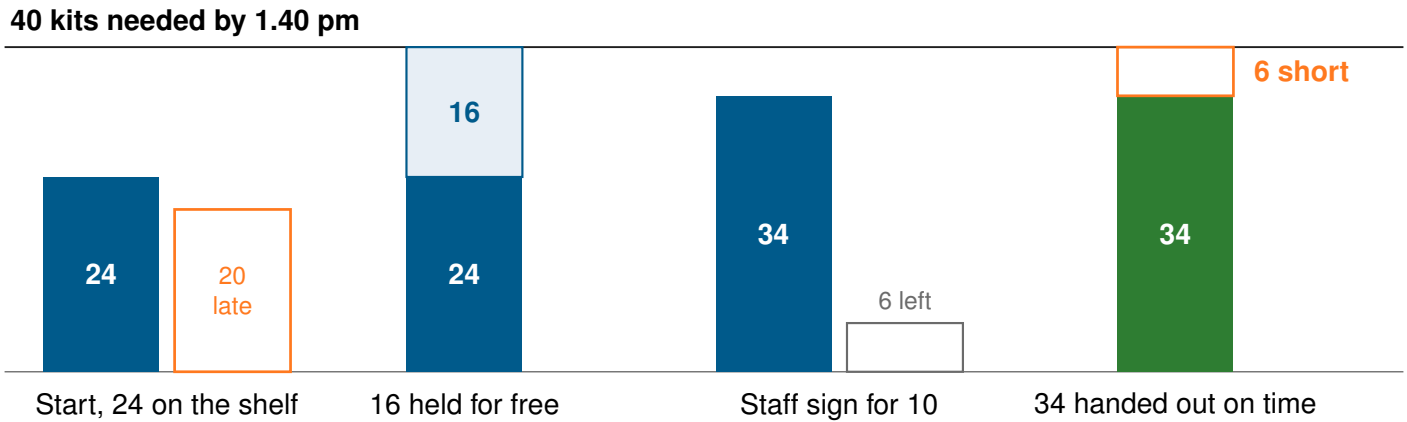
Only Plan B's server can make a change, in every approach tested. In our coordinator the AI picks from the changes that fit.

What if the projector came at 5.45 pm



If the look-up had said 5.45 pm, too late for either slot, Plan B holds the backup projector for free, kept aside for ten minutes and released on its own. Its spending allowance is $0, so it asks the organizer to allow $35. Only after the OK is the backup booked, once. In a recorded run of this case the AI made that choice at 4.48 pm. This is a separate practice case, not part of Jordan's evening.

THE SAME IDEAS AT A MADE-UP FOOD PANTRY



40 kits are needed by 1.40 pm. 24 are on the shelf and 20 are late, so the late ones are not counted until staff sign for them.

Plan B holds a partner's 16 kits for free and asks staff for the handoff. Staff sign for 10.

34 kits are handed out on time. Plan B tells staff 6 will be short, and staff decide what to tell the line. No kit is counted twice.

Plan B counts kits, never people. Who gets food is always a person's call.

Made up, not based on any real pantry's data. Identical boxed kits only.

The earlier 12 case test, and what it found

Runs that met	Campus	Pantry
The fixed rules	6 of 8	4 of 4
The tool-calling model	24 of 24	6 of 12
Our coordinator	18 of 24	9 of 12
The check-all variant	21 of 24	6 of 12

Twelve sealed cases, 120 runs, version 1.1 of the code. Source, the earlier test's notes in the code repository.

The earlier test, on twelve sealed cases, found four problems. A partner's stock that was already checked as unavailable stayed an option, a message about a person running late was not treated as important, the AI was asked again about choices it had already passed on, and a choice refused as out of date cost a new AI call.

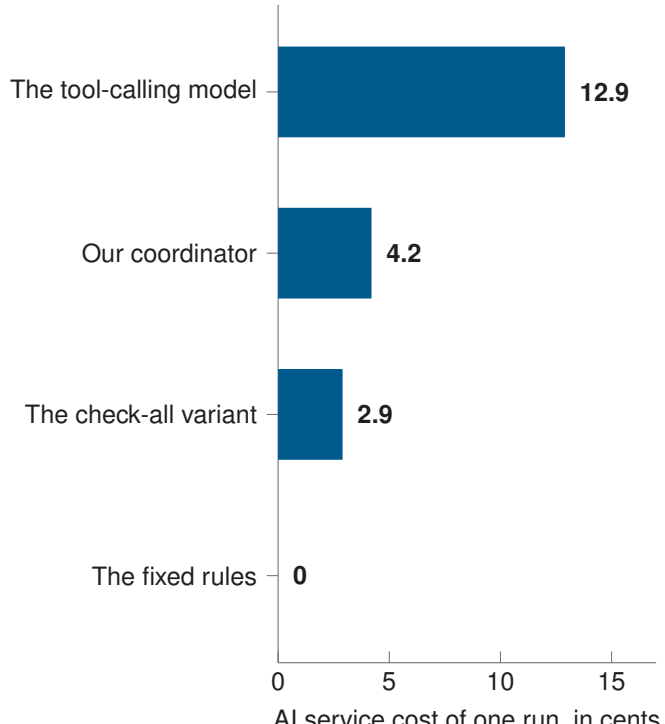
I worked on all four before the final test. A later rerun of the earlier cases still misses two of the cases these changes were aimed at, so not every problem is fully solved.

WHO ELSE DOES THIS, FROM THEIR OWN PAGES

Tool	What its own page says it is for	What Plan B adds for a small group
Sessionboard [1]	AI agenda building, checks agenda changes against rules and sends big changes for sign-off	the one fact checked first, a receipt for every change, free and open
Cvent [2], Bizzabo [3]	agents for event teams, announced in 2026, whose changes a person reviews or approves	runs today on one laptop, free, with no vendor contract
ChatGPT agent, Copilot, Gemini [4, 5, 6]	general assistants that ask before purchases and other actions with real-world consequences	picks only from changes that fit this event's rules
Zapier Agents [7]	automations that stop for approval when the user adds that step	the checks come built in
Group chat, shared sheet, sign-up sheet, campus portal [11, 12, 15, 17]	where the text lands today	works beside them, not instead of them

We have not tested the others. Full sources in the project documentation.

What the final test cost, and the limits per run



All 60 runs cost **$3.61** in AI service fees, with 194 AI calls and no service errors.

Limits per run, at most 48 tool calls, 8 AI calls, $0.50 of AI service fees and 900 seconds. By default the AI service's bill is capped at $20 in total. The final test ran under a $40 cap.

These limits cap the AI service's bill, not event money. Event money always waits for a person's OK.

The same outcomes cost about a third as much when the AI picks from the changes that fit.

Source, the final test's results and spend notes, and the default settings in Plan B's code.

THANKS, AND THE SOURCES CITED IN THIS DECK

Thanks

To the PEAAII organizers at UMass Boston. The AI model is Claude Sonnet 5.5 by Anthropic, reached through OpenRouter. The deck uses the elegant beamer template by Lars Spreng, and the poster the Gemini theme by Anish Athalye with the UMass Boston colors. Plan B is built with FastAPI, uvicorn and Pydantic. Thanks do not imply that any of them endorses Plan B.

Code

github.com/nocdoggo/FA26-COLLIE, free and open source (GPL-3).

Sources

[1] Sessionboard, AI Agenda Builder and Event Scheduler, 2026.
[2] Cvent press release, July 14, 2026.
[3] Bizzabo, Agentic Operating System for Events, October 12, 2026.
[4] OpenAI, Introducing ChatGPT agent.
[5] Microsoft Support, Browse with Copilot.
[6] Google, Manage your tasks and workflows in Gemini Apps.
[7] Zapier Help Center, Add approval steps to your agent's instructions.
[11] Microsoft Support, What is GroupMe.
[12] Google Workspace, Sheets.
[15] UMass Boston, Student Activities.
[17] SignUpGenius, Online Sign Up Sheets.
[28] UMass Boston, Student Groups and Activities, checked October 26, 2026.
[33] OWASP, LLM01 2025 Prompt Injection.
[37] NCSC, Prompt injection is not SQL injection, December 8, 2025.
Full list in the project documentation.